A Drone Swarm-based Wildfire Search and Rescue Method with Autonomous Behavior Modeling and Centralized Task Assignment

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***Abstract*—Forest fires spread rapidly with severe consequences, making timely and effective fire monitoring and personnel rescue crucial. Unmanned Aerial Vehicle (UAV) swarms play a significant role in this scenario. To address the challenge of swarm control, we propose a centralized task assignment method with feedback based on behavior tree and communication link. This approach achieves collaborative task scheduling while preserving individual autonomy. For search behavior, we introduce a heat map search strategy suitable for vertical camera shots, along with optimal parameter selection. Experimental results demonstrate that the proposed swarm control method and optimized search strategy effectively accomplish the task, significantly reducing personnel rescue time by nearly half.**

***Keywords*—Unmanned aerial vehicle (UAV), swarm, behavior tree, task assignment**

I. Introduction

UAVs, known for their cost-effectiveness, compact size, and easy deployment, are increasingly vital in forest firefighting [1]. As outlined in Fig.1, fixed-wing UAVs exhibit high speed and long endurance, enabling fire reconnaissance and monitoring. Quadrotor UAVs, on the other hand, provide hovering and flexibility, crucial for personnel search and tracking. These drones maintain real-time communication with the general commander, reporting information and receiving task assignments. Individual behavior modeling and task assignment are pivotal for efficient rescue operations, requiring a flexible swarm control method. Additionally, the challenges posed by large-scale and high-response rescue missions necessitate the development of an effective joint search strategy.

In the domains of game artificial intelligence and robotics, agents need to construct behavior models that enable autonomous control, allowing them to exhibit intelligent behavior based on environment and task objectives [2]. Finite state machine, a fundamental model for individual behavior modeling, comprises states, transitions, and events, dividing actions into different stages based on states. While intuitive, their structure is challenging to reuse. Decision tree [3], organized as directed trees with nested conditions, deduces actions from top to bottom, achieving modularity but lacking information flow for effective fault handling. Behavior tree [4], with diverse connector and execution nodes, offer varied tree structures. Execution nodes can provide feedback to connector nodes, combining modularity and responsiveness.

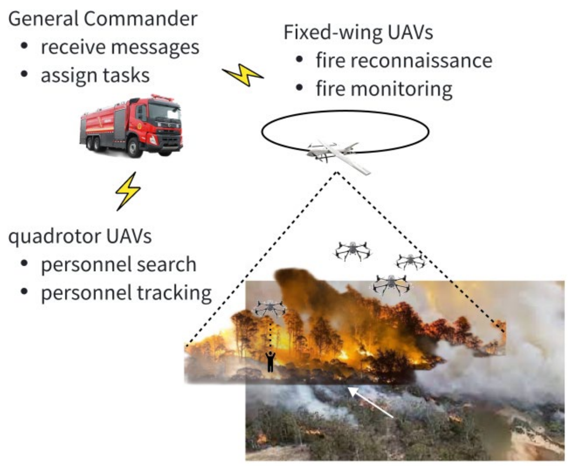


Fig. 1. UAV swarm in forest fire rescue scenario.

UAV swarm control operates on the principle that individual tasks are allocated to each UAV based on user demands, specifying parameters like work area and altitude. UAVs autonomously plan and adjust their flight routes for efficient task completion [5], [6]. Two task assignment methods exist: centralized and distributed [7]. In the centralized approach, a task management unit with comprehensive information decomposes tasks into subtasks, then distribute them to execution units through a communication network [8], [9]. While this structure yields globally or locally optimal solutions, it is computationally complex and lacks execution unit feedback. In contrast, distributed task assignment is execution unit-driven, with the task management unit facilitating information sharing and coordination [10]. Distributed task assignment includes multi-Agent theoretical methods [11]-[13] and market mechanism-like methods [14], [15]. Compared to centralized allocation, the distributed structure offers higher autonomy and robustness, but faces challenges in achieving optimal allocation and is prone to conflicts. Given the limited tasks and swarm size in wildfire scenarios, this paper adopts a centralized structure for task assignment and introduces a feedback mechanism enhancing the flexibility of task execution.

This paper adopts behavior tree for individual drone behavior in wildfire search and rescue scenarios. Task assessment and allocation are centralized under the general commander, allowing individual drones to reject tasks as needed, ensuring efficiency and flexibility. Additionally, a heat map search strategy is proposed for vertical imaging search and rescue, with parameter optimization conducted through simulation experiments.

II. PRELIMINARY

Behavior tree, widely used in gaming and robotics, organizes the behavioral logic of intelligent agents in a tree-like structure. They comprise connector nodes and execution nodes, where connector nodes control the execution sequence, and execution nodes perform condition evaluation and specific action. Common connector nodes include selectors, sequences, and parallel nodes. Selector nodes run the first child node who returns success from left to right, sequence nodes execute all children from left to right until a failure occurs, and parallel nodes run all children simultaneously. Behavior tree starts execution from the root at fixed time intervals, traversing nodes in a depth-first manner and determining the agent's current behavior based on feedback from each node [16].

III. PROPOSED APPROACH

Rescue mission procedures are depicted in Fig.2. This paper discusses a search-and-rescue team comprising a single fixed-wing UAV and multiple quadrotor UAVs. Initially, the general commander commands the fixed-wing UAV to take off from the rear base for reconnaissance. Upon identifying the fire source, it continues to monitor the situation. The commander immediately notifies a quadrotor UAV formation at the front base to conduct personnel search. If endangered individuals are found, UAVs track them, providing accurate information for firefighters. Each base station reserves backup UAVs, with the general commander arranging replacements when the task-executing UAV runs out of battery. If all rotor UAVs are engaged in tracking tasks and the fire area is not completely searched, backup rotor UAVs are deployed until the entire burning area is searched, and all individuals in distress are found.

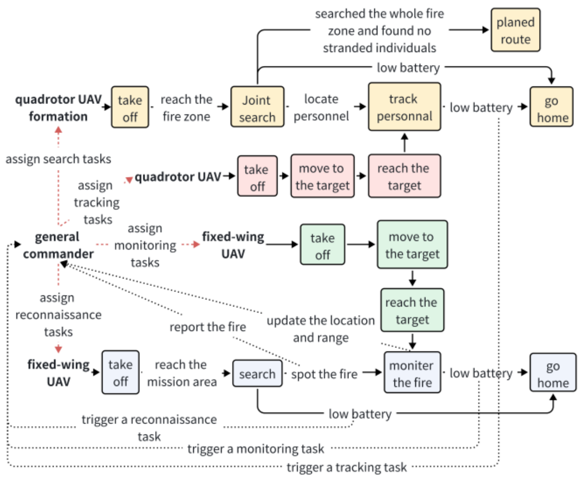


Fig. 2. The flowchart of forest fire rescue task.

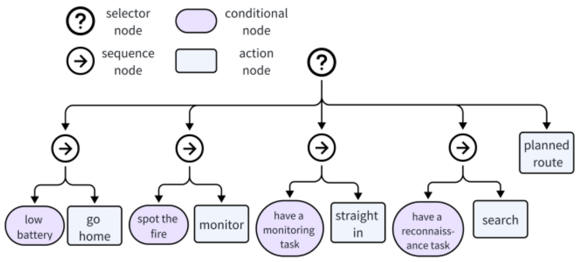
Subsequently, the first subsection will focus on individual behavior modeling, followed by the second subsection outlining the overall task allocation method. The third subsection will discuss a specific strategy for search sub-behaviors.

A. Autonomous Behavior Modeling based on Behavior Tree

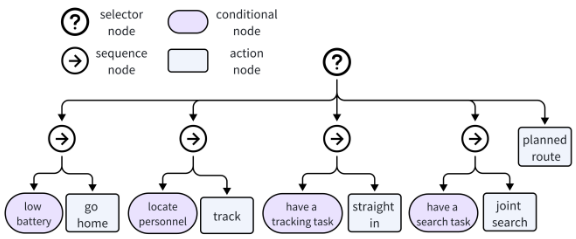
According to the task procedure, fixed-wing UAVs perform search, monitor, straight flight, and return actions, while quadrotor UAVs engage in joint search, track, straight flight, and return. Monitoring involves the UAV circling rapidly over the fire zone, while tracking requires close hovering above individuals for precise location, demanding high accuracy.

We utilize behavior tree to autonomously guide both types of UAVs, ensuring prompt response to commander-assigned tasks and real-world situations. The trees are constructed from top to bottom, and accurately define the requirements for action execution. Additionally, considering the reactive nature of behavior tree, actions requiring urgent attention are placed in the leftmost position under parent nodes, ensuring that UAVs can respond as quickly as possible. For instance, the “go home” node is placed at the far left to ensure UAVs can quickly return to base when low on battery. Routine actions, like “planned route”, are placed on the right for default execution.

The fixed-wing UAV behavior tree is depicted in Fig.3(a). The root node is a selection node connecting to five sub-behaviors: return, monitor, straight flight, search, and cruise. The order of sub-behaviors is crucial, and slight variations may lead to anomaly. Placing monitoring to the left of straight flight is because it is the ultimate goal of reconnaissance. Once a fire is detected, other tasks should give way to monitoring. When the UAV is assigned a monitoring task, to ensure it can continue searching in case it reaches the target point without discovering a fire, the reconnaissance task is also initiated simultaneously. Since search is an alternative in that case, it is placed to the right of straight flight. The quadrotor UAV behavior tree (Fig.3(b)) sequentially evaluates return, track, straight flight, joint search, and cruise conditions until one behavior is permitted. The order of sub-behaviors follows similar considerations to fixed-wing UAVs.



(a) Behavior tree of a fixed-wing UAV.



(b) Behavior tree of a quadrotor UAV.

Fig. 3. Behavior trees of UAV swarm.

B. Centralized Task Assignment with Feedback

Due to the need for coordination among UAV clusters in mountain fire search and rescue, this paper introduces a centralized task assignment method with feedback from subordinates, as depicted in Fig.4. Subordinates refer to all UAVs, controlled by individual behavior trees while providing feedback on demands and task responses to the commander. During the return phase, if a fire point or individuals are identified, UAVs issue monitoring or tracking demands; otherwise, reconnaissance demands are issued. As the search task is collaboratively conducted, the commander uniformly assesses the status of duty UAVs. If no UAV is searching and the fire area is not fully covered, a search demand is proposed. The task generator processes all demands into pending tasks with target points and priority weights. Using the inverse of distance, the evaluator computes the value of each subordinate-task pair. The allocator then produces a task allocation by maximizing the weighted sum of values, communicated to subordinates through the communication link. Subordinates engage in behavior perception; if they are in the search or cruise phase, they accept the task; otherwise, they roll back the task for the commander to reassign to the remaining subordinates.

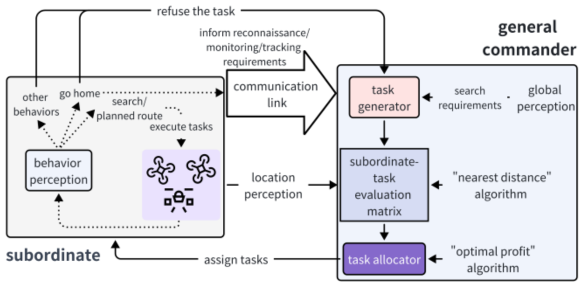


Fig. 4. Centralized task assignment method.

C. Joint Search Strategy

In the behavior tree, the key to each sub-behavior is determining the target point. The search sub-behavior, particularly for quadrotor UAV formations, requires a sophisticated strategy to dynamically determine the target point and complete the whole search within a short time. This paper refers to [17] and proposes a heat map joint search strategy applicable to visual sensors.

In mountain fire search and rescue, all UAVs use a downward vertical camera orientation for target locking. This subsection primarily illustrates the details of the strategy using the example of joint search by a quadrotor UAV formation. The search strategy for fixed-wing UAVs is a special case where the number of UAVs is one. The heat map means gridding the task area, with heat values reflecting the search status of each grid. For quadrotor UAV formations, a virtual node is needed to generate and maintain the heat map and subordinates periodically query this map to adjust flight heading as needed.

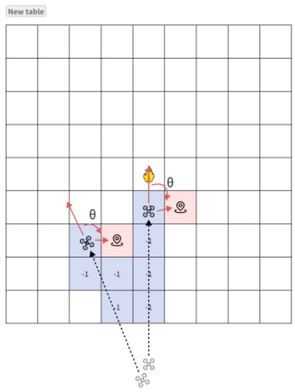


Fig. 5. Heat map joint search strategy.

Specifically, the virtual node will grid the entire fire area based on the reported fire position and range from the fixed-wing UAV. It initializes the heat values of all grids to 1, where each grid approximates the field of view of the quadrotor UAV's camera, as illustrated in the search example in Fig.5. When a quadrotor UAV's camera captures a grid, it updates the grid's heat value to -1. Periodically, each quadrotor UAV queries the closest grid with a heat value of 1 as the target point. The UAV calculates the angle θ between its current heading and the target's relative heading, determining the subsequent heading α. To maintain flight stability, the heading α is constrained by the UAV's maximum turning angle β. The real-time calculation formula for the heading is as follows:

$$\alpha = \begin{cases} \hat{\alpha} & \text{if } |\theta| \leq \Theta \text{ and } n < N \\ \hat{\alpha} + \beta & \text{if } |\theta| \leq \Theta \text{ and } n \geq N \\ \hat{\alpha} + \theta & \text{otherwise} \end{cases} \quad (1)$$

In the formula, $\hat{\alpha}$ is the UAV's previous heading, Θ is the turning angle threshold. If θ is less than this threshold, it implies the current heading is approximately correct, avoiding unnecessary turns to save search time. n represents the number of times the UAV maintains its heading, and N limits this count. After multiple heading maintenance cycles, the UAV is compelled to turn with the maximum angle, enhancing the exploration of unknown areas in different directions. The task concludes when all grid values within the fire area are -1.

IV. EXPERIMENTAL RESULT

All experiments were conducted on a dedicated simulation platform using the platform's script language to define logic for sub-behaviors and construct behavior trees to control UAVs. We initially established the entire task scenario and verified in real-time simulation that the proposed autonomous behavior modeling and centralized task assignment method could achieve mountain fire search and rescue by a UAV cluster. Then, we simplified the task scenario to a joint search scenario involving quadrotor UAVs, and optimize parameters for the heat map search strategy proposed in Section III-C.

The simplified task scenario assumes that the fixed-wing UAV has already identified a mountain fire. The general commander instructs the quadrotor UAV formation to proceed from the front base to the fire area for search tasks. The task is considered complete when all trapped individuals are detected, and the entire fire area is thoroughly searched. We jointly optimized the search strategy's parameters Θ and N based on the completion time of the task, and considered the time it takes to discover the last trapped individual to ensure all tasks reach their objectives. The simulation scenario is illustrated in Fig.6. In the experiments, the parameters for the fixed-wing UAV are based on the Delair UX11 drone, a professional-grade fixed-wing UAV with long flight endurance and the capability to equip a thermal imaging sensor, commonly used for monitoring, mapping, and search and rescue applications. The parameters for the quadrotor UAV are based on the DJI Mavic 2 Enterprise Dual, a compact and versatile drone designed for professional emergency services. Experimental parameters are detailed in Table I.

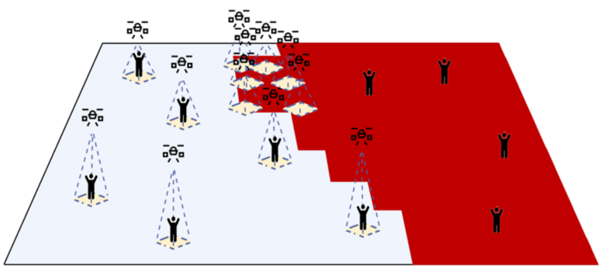


Fig. 6. Simulation scenario.

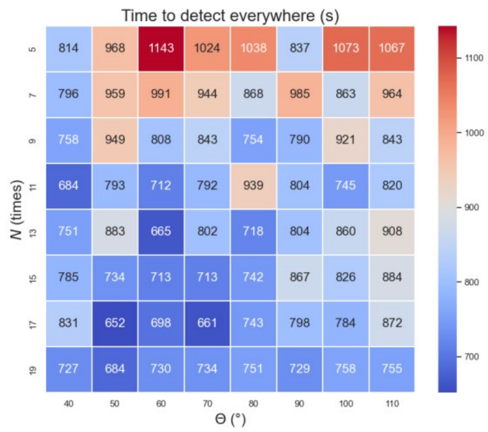
TABLE I. LIST OF COMMON HYPER-PARAMETERS

Parameter		Value
the range of the fire area		4.16 km^2
the number of trapped individuals		10
area of each grid		14400 m^2
quadrotor UAVs	quantity	12
	horizontal field of view of the camera	60°
	vertical field of view of the camera	60°
	flight height (relative to the ground)	100 m
	speed	50 km/h
	maximum turning angle β	120°

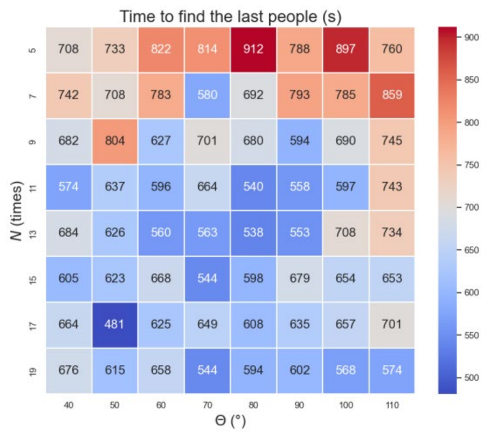
The maximum non-turning count N in the search strategy enhances the exploratory nature of UAVs, while the turning angle threshold Θ maintains stability in search heading. The choice of both parameters significantly impacts the efficiency of the strategy. In this experiment, N is varied from 5 to 19 with a step size of 2, and Θ is varied from 40° to 90° with a step size of 10°.

Fig.7(a) illustrates the variation in task completion time with changes in these two parameters. Smaller values of Θ make UAVs more prone to changing their heading. When the difference between the target and current heading is small, turning may not significantly improve the search direction, but will affect the overall search progress. Conversely, setting Θ too large can result in delayed correction when UAVs are oriented unfavorably for exploring new areas, leading to redundant searches in old regions. Smaller values of N make UAV behavior more random, fostering exploration of different regions. An appropriately chosen N contributes to long-term task execution efficiency. The experimental results indicate that the maximum non-turning count N has a more significant impact on search duration, and larger values lead to faster rescue. This suggests that excessive randomness can disrupt the original search plan. Smaller Θ values result in better search efficiency, highlighting the need for timely correction when the heading deviates from the optimum. The optimal search duration for the entire fire area is achieved at $\Theta = 50°, N = 17$, reducing nearly half of the longest time shown in the graph.

Fig.7(b) illustrates the time taken to discover the last trapped individual with varying parameters. Under all parameter settings, the joint search strategy based on the heat map successfully discovers all trapped individuals. The trends align closely with Fig.7(a), and the optimal performance is consistently achieved at $\Theta = 50°, N = 17$, further confirming the effectiveness of parameter optimization.



(a) Time to detect everywhere.



(b) Time to find the last people.

Fig. 7. Heat map for parameter optimization.

V. Conclusion

This paper employs behavior trees to achieve personalized behavior modeling for different types of UAVs, enabling them to act based on tasks assigned and environmental factors. Simultaneously, task allocation problem in UAV cluster mountain fire search and rescue is addressed primarily through a centralized approach. The introduction of a feedback mechanism allows subordinates to retreat from tasks based on real-world conditions, with reassignment being carried out by the commander. Lastly, a heat map-based search strategy is proposed for UAV search behavior, striking a balance between exploration and exploitation, achieving promising search and rescue outcomes in experiments.

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